

example: will do Leb. decomposition of μ w.r.t γ and find RN deriv of μ_a w.r.t γ , where

$$\mu = \text{Cauchy}(0,1), \quad \gamma = \chi^2(n) \equiv \Gamma\left(\frac{1}{2}, \frac{n}{2}\right) \quad \left(\equiv \text{Gamma}\left(\frac{1}{2}, \frac{n}{2}\right) \right)$$

Since γ has support $(0, \infty)$, will define

$$\mu_s(A) = \mu(A \cap (-\infty, 0]), \text{ then } \mu_s \perp \gamma \quad (\text{can use } B = (0, \infty) \text{ here})$$

$$\begin{aligned} \text{Choose, } \mu_a(A) &= \mu(A) - \mu_s(A) \\ &= \mu(A) - \mu(A \cap (-\infty, 0]) \\ &= \mu(A \cap (0, \infty)) \end{aligned}$$

→ Hence $\mu = \mu_s + \mu_a$ is true. Also $\mu_s \perp \gamma$. Need to check this $\mu_a \ll \gamma$ and find $\frac{d\mu_a}{d\gamma}$:

(method 1:)

First note that if m' is the Leb meas on $(0, \infty)$, i.e. $m'(A) = m(A \cap (0, \infty))$, m being the usual Leb meas, then

$$\gamma(A) = \int_A \frac{e^{-x/2} x^{n/2-1}}{2^{n/2} \Gamma(n/2)} dm'(x) \quad [\text{note: we do not need } I_{(0, \infty)} \text{ here since we are working with } m' \text{ and not } m]$$

$$\left(\frac{d\gamma}{dm'} \right)(x) = \frac{e^{-x/2} x^{n/2-1}}{2^{n/2} \Gamma(n/2)} > 0 \text{ a.e. } (m')$$

$$\Rightarrow \left(\frac{dm'}{d\gamma} \right)(x) = \frac{2^{n/2} \Gamma(n/2)}{e^{-x/2} x^{n/2-1}} \text{ a.e. } (\gamma) \quad (\text{By proposition 4.1.2 (iii)})$$

↳ also true a.e. (m') , since by $\nearrow$, $m' \sim \gamma$.

$$\text{Also } \mu_a(A) = \mu(A \cap (0, \infty))$$

$$= \int_A \frac{1}{\pi(1+x^2)} I_{(0, \infty)}(x) dm(x) = \int_A \frac{1}{\pi(1+x^2)} dm'(x)$$

do not need the indicator, since dealing with m' and not m . (can verify).

$$\text{i.e. } \left(\frac{d\mu_a}{dm'} \right)(x) = \frac{1}{\pi(1+x^2)} \text{ a.e. } (m') \quad \dots \textcircled{2}$$

$$\textcircled{1} + \textcircled{2} \text{ with prop 4.1.2 (i) gives } \left(\frac{d\mu_a}{d\gamma} \right)(x) = \frac{1}{\pi(1+x^2)} \frac{2^{n/2} \Gamma(n/2)}{e^{-x/2} x^{n/2-1}} \text{ a.e. } (\gamma) \quad (\mu_a \ll \gamma \text{ and } \mu_a \ll \gamma)$$

(method 2)

$$\mu_A(A) = \int_{A \cap (0, \infty)} \frac{1}{\pi(1+x^2)} dm(x)$$

$$= \int_A \frac{1}{\pi(1+x^2)} dm'(x) \quad , \text{ where } m'(A) = m(A \cap (0, \infty))$$

$$= \int_A \frac{1}{\pi(1+x^2)} \frac{[e^{-x/2} \cdot x^{n/2-1} / 2^{n/2} \Gamma(n/2)]}{[e^{-x/2} \cdot x^{n/2-1} / 2^{n/2} \Gamma(n/2)]} dm'(x)$$

$\chi^2_{(n)}$ "density" on $(0, \infty)$
 (do not need an indicator $I_{(0, \infty)}(x)$ since working with m')

$$= \int_A \frac{1}{\pi(1+x^2)} d\nu(x)$$

since $\nu(A) = \int_A [e^{-x/2} \cdot x^{n/2-1} / 2^{n/2} \Gamma(n/2)] dm'(x)$
 and in that case for any f (ν -integrable),

$$\int f d\nu = \int f(x) [e^{-x/2} \cdot x^{n/2-1} / 2^{n/2} \Gamma(n/2)] dm'(x)$$

So $\mu_A(A) = \int_A h(x) d\nu(x)$ where $h(x) = \frac{1}{\pi(1+x^2)} \frac{2^{n/2} \Gamma(n/2)}{e^{-x/2} x^{n/2-1}}$ a.e. (m')

So $\mu_A \ll \nu$ and $\left(\frac{d\mu_A}{d\nu}\right)(x) = h(x)$

Note: you can also have $h'(x) = h(x) I_{(0, \infty)}(x)$ as $\frac{d\mu_A}{d\nu}$

since $\{x : h'(x) \neq h(x)\} \subseteq (-\infty, 0]$ and $\nu(-\infty, 0] = 0$.

So $h' = h$ a.e. (ν). Similarly, other equivalent (a.e. (ν)) forms of $(d\mu_A/d\nu)$ are possible!